

SHENG SHANG, China

WMO: 584730

Lat: 30.722N

Lon: 122.818E

Elev: 197

StdP: 98.98

Time Zone: 8.00 (E08)

Period: 82-90

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.3	1.1	-8.9	1.8	1.6	-7.2	2.1	2.4	18.1	4.7	17.0	3.7	9.8	340	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.8	29.0	26.2	28.2	25.9	27.5	25.6	26.7	28.1	26.2	27.5	26.0	27.1	4.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	22.2	27.5	25.9	21.7	27.1	25.6	21.3	26.8	85.1	28.3	83.3	27.8	81.9	27.2	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	15.9	14.3	-2.0	31.0	1.6	1.0	-3.2	31.8	-4.1	32.4	-5.0	33.0	-6.2	33.7		
(5)			-4.1	27.7	1.3	0.8	-5.0	28.2	-5.8	28.6	-6.5	29.1	-7.4	29.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	6.0	5.5	8.0	12.0	16.8	21.1	25.1	26.1	23.1	19.3	14.7	8.8		
	DBStd	7.56	2.87	2.57	2.74	2.42	2.23	2.09	1.70	1.29	2.01	2.11	3.35	3.51		
	HDD10.0	406	128	127	74	8	0	0	0	0	0	0	5	65		
	HDD18.3	1737	383	359	319	190	58	2	0	0	0	15	115	296		
	CDD10.0	2450	3	1	13	68	211	334	468	500	392	287	146	27		
	CDD18.3	739	0	0	0	0	10	87	210	242	142	43	6	0		
	CDH23.3	3962	0	0	0	0	1	134	1354	1956	491	23	2	0		
	CDH26.7	390	0	0	0	0	0	6	132	231	21	1	1	0		
(14) Wind	WSAvg	7.2	7.6	8.0	7.7	7.2	7.0	6.6	7.5	6.6	6.6	6.8	7.3	7.4		
(15) Precipitation	PrecAvg	1214	66	72	105	94	108	190	101	144	118	79	73	64		
	PrecMax	1531	213	150	246	196	231	395	335	371	263	252	188	182		
	PrecMin	668	14	16	23	26	25	9	0	7	15	1	14	0		
	PrecStd	209	38	39	65	36	48	95	80	97	71	65	50	48		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	13.7	16.9	18.9	22.6	27.3	30.6	30.6	28.3	25.4	21.5	18.1		
		MCWB	12.8	12.8	13.8	14.7	20.3	24.1	26.2	26.9	25.6	22.4	19.4	15.1		
	2%	DB	12.8	11.6	14.9	17.6	21.2	25.7	28.8	29.4	27.2	23.8	20.4	16.3		
		MCWB	11.9	10.3	13.5	15.1	19.5	24.2	26.2	26.4	25.1	22.2	19.0	13.6		
	5%	DB	11.4	10.5	12.9	16.5	20.4	24.8	28.0	28.6	26.5	22.7	19.7	14.7		
		MCWB	10.0	9.4	11.5	15.1	19.0	23.9	25.9	26.1	24.7	20.7	18.2	12.4		
	10%	DB	10.2	9.3	11.6	15.6	19.8	24.0	27.2	27.8	25.9	21.9	18.8	13.3		
		MCWB	8.4	8.0	10.3	14.3	18.6	23.3	25.6	25.8	24.4	19.4	16.8	11.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	12.9	15.4	17.1	21.3	25.3	27.1	27.7	26.2	24.0	20.3	16.0		
		MCDB	13.6	13.5	16.0	17.5	21.9	26.1	28.8	29.8	27.2	24.8	21.0	17.7		
	2%	WB	12.0	10.8	13.6	16.2	20.4	24.6	26.6	26.8	25.7	23.0	19.5	14.1		
		MCDB	12.6	11.2	14.3	16.8	20.7	25.2	27.9	28.3	26.6	23.5	20.2	15.6		
	5%	WB	10.4	9.6	12.0	15.5	19.7	24.1	26.2	26.4	25.3	21.5	18.7	12.8		
		MCDB	11.4	10.3	12.7	16.1	20.1	24.6	27.3	27.8	26.2	22.2	19.5	14.3		
	10%	WB	8.7	8.3	10.7	14.7	19.1	23.4	25.9	26.1	24.7	20.0	17.5	11.7		
		MCDB	10.0	9.0	11.4	15.3	19.6	23.9	26.9	27.4	25.6	21.5	18.5	13.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.0	2.7	3.3	3.4	3.3	2.8	2.9	2.8	2.7	2.6	3.0	3.3		
		MCDBR	4.4	4.7	5.7	4.8	4.3	3.8	4.0	3.5	3.5	3.5	3.5	5.2		
	5% WB	MCWBR	4.4	4.3	4.0	3.7	3.3	2.4	2.5	1.7	2.4	2.7	3.3	4.3		
		MCWBR	4.1	4.5	5.2	4.2	3.4	3.3	3.8	3.2	2.7	2.6	3.2	4.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.528	0.566	0.606	0.611	0.593	0.634	0.594	0.564	0.530	0.507	0.523	0.512		
		taud	1.811	1.756	1.703	1.750	1.838	1.819	1.944	2.023	2.057	2.000	1.884	1.843		
	Ebn at Noon	Ebn at Noon	664	686	696	715	732	700	726	743	752	733	666	651		
		Edh at Noon	178	206	232	229	210	214	188	172	160	159	164	165		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.01	2.54	3.63	4.28	4.68	4.17	5.68	5.47	4.32	3.51	2.39	1.96		
	RadStd	RadStd	0.27	0.50	0.30	0.46	0.51	0.56	0.67	0.59	0.44	0.43	0.31	0.30		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+75	N/A

Nomenclature: See separate page